All right, good afternoon. We'll go ahead and start. I want to first just do a quick review of what we talked about last time, which was also a review. But we just kind of went back over some of the most important concepts from probability, the prerequisite class. So we started with just some basic definitions here. We talked about our conditional probability formula here, right, which we extended a bit to get Bay's rule. The Bay's rule, very important rule and probability.

We're going to talk about Bayesian statistics, the Bayesian. Bayesian paradigm for doing statistical inference, which is sort of an extension of Bayes rule itself.

So extending this idea of updating a probability to the idea of updating an entire probability distribution.

We'll talk about that starting today. We've got random variables and then probability distributions, which are the tools we use for working with random variables. Right. So we need to know if we're working with a discrete random variable or a continuous random variable. If we're in the discrete case, we'll often work with a PMF or a probability mass function, right, just the function that tells us the probability that the random variable big X takes on some specific value, little X. If we're in the continuous case, we have the PDF, the probability density function, which is the analogous case for continuous random variable. So now we can integrate this PDF, integrate little F of X, to get a probability of a random variable falling within some interval, falling between lower bound A and upper bound B. And then for both of these, we have the CDF, the cumulative. cumulative distribution function, right, which is just the probability that the random variable x takes any value all the way up to some value little x. Should be familiar with expectations, right, in both the continuous and the discrete case, how to find the expectation or the mean of a random variable, and also the variance, which is a specific type of expectation, right, the expectation of this squared difference between the random variable and its mean. And then we looked at this table, this table of important probability distributions, right? So be familiar with all of these distributions. We'll work with all of them throughout the corridor. As you're working with or thinking about a specific probability distribution, keep in mind some things like what are the values that the data can take, right? Is it discrete, like in the Poisson case? You can take on values zero, one, two, the counting numbers.

Is it continuous like the normal distribution, right? Can it take on values from negative infinity to infinity or some other, right? Continuous interval. maybe zero to infinity for the gamma, right? So one thing to be aware of is what values our data can take, what values X can take, but then also parameter restrictions. What values can the parameters themselves take? In a binomial distribution, we have this parameter P, and it's a probability, right? So it's restricted to be between zero and one. So we might have restrictions on what values the data can take, X, we might also have restrictions on what values the parameters. So these parameter restrictions will come back up again today. We'll talk about that. So be aware of these restrictions, be familiar with these distributions, and giving you the means and the variance is here. So you can reference this table if you want to quickly get the mean or the variance for one of these specific distributions. But we also talked about expectation and variance. So be familiar with how these are calculated as well, be able to calculate a mean or variance if it's not just given to you. So you could go and calculate any of these values in these two columns here. talked about a couple important results from probability, right, the law of large numbers, which just tells us that our sample mean x bar converges to the true expectation of x as our sample size grows, and then the central limit theorem, which talks about the distribution of x bar, and so it says under the right conditions, regardless of what distribution we have for the individual data points x, right, binomial, poisson, gamma, whatever, the distribution of their sample mean x bar we'll approach a normal distribution, right? So we can approximate it with a normal distribution if we have a reasonable sample size. And then we ended just by

talking about what is statistical inference, how is it different from probability theory? Right, your prerequisite course. So we said in probability, you're starting with known parameter values for some distribution, right?

In the most simple case, you know the probability of heads for a certain coin, right? And you're asked to compute probabilities of events or data, probabilities of X with those known parameters.

And in statistics we flip that on its head, we do the inverse, right? So we start with some data, start with some observations, some values of X that we've observed from some random probability distribution, right? And then we want to make some inference about those unknown parameters. We don't know the parameters, we observe data, we'd like to infer something about those parameters.

That's the goal of statistics and gave some examples there and then just kind of ended some distinctions between parameter parameter space right a statistic which is just a function of the data not the parameters some examples the sample mean the sample variance and then just different tasks that we might do with statistics some tasks things like prediction or estimation or hypothesis testing okay so today we're going to go over the lecture two notes, talk about frequentist and Bayesian inference, the two different paradigms for doing statistical inference that we'll talk about in this course, the two main ways. We'll talk about the difference, and then we'll kind of dive in a little bit to the Bayesian side of things. We'll stick with Bayesian statistics for the next couple lectures. Eventually, we will come back to frequent statistics. So what is statistics? We said it's this idea that we have some observations, some data, and we'd like to infer something about the parameters of a probability distribution, right?

So we need some definitions here. So first we're going to define a population as a full set of units that we're interested in studying. You know, the collection of something that we'd like to learn about, right? Maybe if you're interested in studying people, it's all the people in California. You might like to know something about the collection of people in California. Or maybe it's a species of bacteria, right, or bird or something else.

If you're in business, you might be interested in transactions. So maybe all the customer transactions at a company like Amazon, that could be a population, right? A set of units that we're interested in studying. Or maybe it's all the possible hands in poker. Generally, a population will be very large. It'll have a lot of units within the set. All the possible hands of poker. There's a lot of them. All the people in California, there's a lot of them. So it might be difficult to learn something about this population. You'd have to know something about very, very many individual units in that population. And so often instead, we work with a sample. We don't get the full population. If we had the full population, if we measured something for everyone in the state of California, we'd kind of have our answer. There'd be no need for statistics. But instead, we work with a sample. And so a sample is some subset of those units in the population. And we only get measurements or data for that subset. It's kind of the main idea of statistics. We only get the subset of data, the sample, but we'd like to infer something about the population, or specifically some parameter, parameters in that population.

And a sample is typically random. You might get a different sample or a different sample if you continued this sampling procedure. There's randomness or uncertainty, which leads to the idea of modeling things with probability distribution.

So some examples of samples, right? If you're studying people in California, you might just take a random sample of a thousand people in California.

Maybe you can't ask everybody in the state a question, but you can't ask a thousand people, right? You can collect a sample. Or maybe you have the bacteria

that are contained on a single petri dish, right? That's a sample from this population of all, or the entire species of bacteria.

You might have all the customer transactions at Amazon for a small window of time, like an hour, right? Or maybe you observe 10 hands of poker at a casino. So some subset of this population is what we get to observe, and we're interested in a parameter.

So a parameter is some numerical measurement that describes some characteristic of a population.

Very often, this is the thing we would like to know.

Maybe we want to know the proportion of California residents with a college degree. We had access to the full population, right? If we knew for every single person in the state of California, whether or not they had a college degree, we'd have our answer. There's no work to be done, no randomness, but instead we're going to work with a sample to infer about this proportion of California residents with a college degree, or the average mass of bacteria from some specific species, or the range of Amazon transactions amount, a measure of spread or uncertainty, or the probability of a full house or a specific type of hand in the game of poker. So we defined a parameter here. We also talked about probability distributions, right, recapping from your probability course. So very often we're going to choose a specific probability distribution to model some population, right? And when we talk about a probability distribution, they have parameters. We talked about the parameter P for the binomial distribution, right? And so the parameters associated with the distribution that we choose for our model, they may or may not be the actual parameters that we care about. For example, if we're talking about modeling IQ, it's known that IQ follows a bell shape. So, right, you might assume a normal distribution if you're modeling IQ, right? And one example that you might be interested in is the mean or the average IQ in the population. And if you look at, yeah, question? Yeah, so there's different ways or strategies to choosing a model. But things you should think about is what is the support of the data, what values can the data take, any restrictions, things like that. You can look very often you'll want to plot your data, look at the shape of it, so you does it kind of support one probability model over another. Yeah, so that's a little bit the art of statistics is learning how to choose an appropriate probability distribution. But say in IQ, the case of IQ, a lot of people have worked with IQ, it's kind of meant to be normally distributed. So normal is going to be a good probability distribution for this case. And so one thing you might be interested in is the mean or the average IQ in the population. And if we're modeling it as normal, the mean mu is a parameter of that normal distribution, right? So the thing we're interested in is directly parameterized by the model that we chose. So that can happen, but it doesn't always have to happen. So for example, you might be interested in something like the probability that an IQ is greater than 140. That's a characteristic of this population, right, all IQs. It's a characteristic of that, but it's not directly a parameter of the normal distribution. Now, if you model IQ well using a normal distribution, you can still give an answer to that question, right, but it's not directly parameterized by the model. So just keep it out in mind. It doesn't necessarily have to be the same as the parameter from the probability distribution that you're working with. So I want to just kind of visualize a sample here and a population, right? So looking at this first plot, we have a bunch of circles here, right? And collectively, these circles can make up some population, right? And maybe you have the resources to go and and sample all of these circles, right, and you'd have your answer, and there's no uncertainty, but typically that's not feasible. We don't have the resources, or we don't have the time to go and collect data on every single population unit, right, especially if we're talking about very large populations. So instead, we work with a sample. So all of these blue circles are part of our sample, randomly selected sample here. And maybe we do have the resources to go and collect data for that sample. So that's the idea. And if you repeated that sampling procedure, you might get different blue circles. And if you did it again, you might get different blue circles. So

there's uncertainty there, right? There's randomness involved in that sampling process. There's no randomness involved at the population level. The randomness comes from the sampling here. And now maybe there's measurements associated with each of these. So we have numbers here, four, six, one, six, and so on. So one parameter that you might be interested in is the population mean. Right? If you added up all the values in these circles and then divided by the number of circles that there are, that would tell you the population mean, right? In this case, it's 4.3. 4.3 is the population mean, right? It's a parameter that we might care about. And again, if we did have all of this data for every single circle here, every unit in the population, there'd be no uncertainty. We'd have the answer. It's 4.3. There's no randomness involved. There's nothing to model. It's just 4.3. But instead, we work with a sample. We just get these blue values. We don't get to observe the rest, so we don't get to observe the fact that the population average is 4.3. But maybe working with these blue circles, we say, okay, what if we just take the average or the mean of these blue circles? That's probably a good guess for the average of the population. In this case, it's 4. Pretty close to 4.3. It is a pretty good guess here. That's kind of the idea behind sampling. Hypothetically, we could go and sample all of these points. but it's not practical. So instead, we get some subsample the blue dots here. In that case, in these two plots, we had a finite population. There's a finite number of circles in the population. That doesn't have to be the case. You could have an infinite population. So imagine these rows and columns kind of going on forever. There's no limit to the number of population units here, right? Maybe if you're flipping a coin, thinking about the probability, if the coin comes up heads, you could flip that coin forever and ever and ever. There's no limit to how many times you can flip it. So your population there is kind of the infinite sequence of coin flips, if you will. It's an infinite population. You can still take a sample. You could do 100 coin flips, take some sample from that infinite population, and maybe in some hypothetical world where you did have infinite time and infinite resources, then you could sample all of those. But in the real world, that's not the case. we're going to have to work with some finite sample. Yeah, question. You can have a population parameter, right, as the, say, the probability that the coin comes up heads, right? Probability you want to just defined as kind of a limit of the heads occurring, a number of heads occurring over the total number of flips. So it does exist, right? Hypothetically, you could observe it if you sampled all of these, but in the real world you can't. So there's no practical way to observe that population parameter, but we can still estimate it from sample data. So this distinction between the population and the sample is very important, right? It can lead to a lot of confusion when learning statistics if it isn't clear in your mind. So whenever you're looking at a problem, try to very clearly distinguish what is the population, what is the parameter from that population that we care about, what's our sample, and then anything we calculate using the sample, things like statistics, the sample mean, right? Those are typically random because the sample itself is random. So keep those things clear in your mind. Keep in mind that the source of randomness typically comes from the sample, and I think that will help quite a bit as you're learning these topics. So as we do statistics, there's different paradigms, different ways that people do statistics. At its core, it's always about this idea of learning about some population parameter or inferring about a parameter based on data just from a sample. But there's different ways that people might do this. And the two major philosophical viewpoints, are the frequentest and Bayesian approaches. Kind of interesting, our statistics department here has kind of a long history of specializing in Bayesian statistics, which is not really the norm, especially, say, 10 years ago. It's becoming a little more popular now with a lot of the recent work in AI and machine learning, but especially 10 years ago, there wasn't as much Bayesian statistics. It was more of a specialty topic. At a lot of departments, right, it's kind of taught as an elective at the graduate level, an undergraduate would never get to see it. So you're kind of lucky. You do get to see Bayesian statistics at the undergrad level here. That's kind of unique. So we'll talk about the difference between these two. Starting with frequentist statistics, right, which kind of aligns with the notion of probability that you define in your probability theory class. So in the frequentist approach to statistics, when we think about parameters, right, we treat them as some fixed but unknown

value. So going back to this plot here, right? 4.3 was the unobserved population average. It's fixed, meaning it's not random. There's nothing random about it. It's just 4.3. It's just that constant, right? But it's unknown. It's a fixed unknown value. And that's how frequentists treat or deal with parameters, particularly parameters in their probability distribution that they're modeling with. So because this parameter is fixed, it's not random, right? We can't make probabilistic statements about the parameter. So when we talk about interval estimates, confidence intervals, if you've heard the term, right? We can't talk about the probability of a parameter being in some calculated confidence interval, right? If you calculate a confidence interval from 5 to 10, you can't talk about the probability of the true parameter being in that interval because the true parameter is not random, right? It's either some fixed value outside of the interval, in which case the probability that it's in the interval is zero, or it's some fixed value inside the interval, in which case the probability that it's in the interval is 1, right? We don't talk about the probability, or we can't meaningfully talk about probability of a parameter being inside a specific interval in this frequentist approach, because we think of parameters as fixed but unknown, right? Typically, we don't know if it's in the interval or out of the interval. So instead we talk about confidence, right, which comes from this idea of long run or asymptotic results, right? So maybe if we were to repeat our sampling procedure, right, randomness is coming from the sampling procedure, right? So if we were to repeat it, we might get a different interval, maybe 6, 10, or 7 to 11, right? If we repeat this many times, we're going to get many different intervals. So in the long run, over the sampling procedure, the intervals are kind of random, but for any given interval, we can't talk about the probability that the parameter falls with an interval, but maybe we can think about this asymptotic idea, this long run idea of if we repeated our sampling procedure many, many times, couldn't really do it in practice, but if we could, you know, what proportion of those intervals would contain the true value, right? And that's this idea of confidence. We'll talk a lot more about this towards the end of the course. But again, just trying to get this idea across that frequentist statistics treats these parameters as fixed but unknown quantities.

Bayesian's don't do that.

They treat random variables as random, or sorry, they treat population parameters as random variables.

So they do have probability distributions. Regardless of whether or not we think that it's some fixed unknown value that we'd like to learn about, we deal with them or we treat with them as random variables having probability distributions, right? And so our interpretation of probability is a little bit different in this case. It's not this long-run asymptotic idea of probability. It's reflecting our own uncertainty about the parameter at any given state of time. Yeah, question. The parameter could or could not change over time, right? It works in both cases. Just like in this case where it is some fixed but unknown thing, we can still be basian.

We can still use probability distributions to model this thing that's unobserved, right, even if we don't actually believe it's a random thing.

And we do that through this extension of Bayes rule that I alluded to earlier. So that's what we're going to start talking about.

But even if the parameter itself isn't changing, right, our knowledge of the parameter could change. Before we collect any data, we might have some knowledge of the parameter. I work with a lot of soil scientists and ocean scientists. And they have a lot of knowledge about their field, right? They have kind of this prior information, right? And so I could rely on that information as I'm estimating some parameter.

That's this idea of a prior, right? It's our uncertainty about the parameter before seeing any data. But at some point, we're going to take a sample. We're

going to get data, and that data is going to tell us something about the parameter. So we have more knowledge about the parameter, less uncertainty about the parameter. We could update our beliefs, which results in something called the posterior distribution. So moving from this prior distribution to the posterior distribution, that's the Bayesian framework. But at any given state of time, we have a probability distribution that encapsulates our belief about the parameter that we're interested in. There's some hands up? Why isn't it more? So good question. There's sort of a long history. There's a feud between Bayesian and Frequentists for a long time, not so much anymore. But one thing that a lot of frequentists complain about is that there's subjectivity when you're Bayesian, right? Because your priors might not be the same as my prior, meaning your posterior might not be the same as my posterior, right? And so they like this idea that if two people were to do the same analysis, they'd get the same results, but maybe that's not the case in the Bayesian setting. So that's one criticism of Bayesian modeling. Now you can construct priors that don't really, you know, inject your own beliefs into them. You can construct things like a flat prior, say if you're modeling a probability, a uniform prior. It doesn't favor any one value over another. So there are kind of ways or arguments around that, but generally that's a big criticism of Bayesian modeling.

Yeah, that's a great observation. Under the right conditions, yeah, as your sample size grows, your posterior will do what's called contract around the true population parameters.

That's an asymptotic argument, right? And in the real world, we only have a finite sample size, right? So it's a nice property. Does it always apply to the problem we're dealing with?

That's a little less unknown. So in this case, this is a fixed but unknown parameter, right?

But we could still model it as a random variable. A frequentist wouldn't do that, but if you're a Bayesian, you could do that. We could say, okay, we don't know what this value is, but let's put a normal prior on it centered around five with a standard deviation of five. So pretty spread out, but we think five is a reasonable guess, right? That's a prior distribution we could use for this unobserved population average. So we're treating it as a random variable. Random variables have probability distributions, and that prior distribution is our belief about the parameter before we collect any data. And then we'll go and we'll collect these sample points and we'll update our prior distribution. We'll talk about how to do that. We haven't gotten there yet. We'll talk about that we can update our prior distribution, get a posterior distribution, right? And hopefully that posterior distribution is a little better, a little closer to the truth, a little less uncertain than our prior, right, after we've observed the data.

So it doesn't really matter whether you think a parameter is fixed but unknown. And in my view, pretty much all parameters are fixed but unknown, but I will model them as random variables. So the, yeah, the two are kind of distinct. Two what? Interchange frequentist and Bayesian paradigms? Yeah, I mean, lots of people do that. They'll, you know, for one problem, they might use a frequentist approach. For another problem, they might use a Bayesian approach. There's nothing that says you have to fully be in one camp or the other, right? You could do what's best for the problem at hand. As you get into very complex models, you'll find that, you know, not in this course, but, you know, in graduate level courses or if you, you know, really dive deep into statistical modeling. Bayesian approach has a lot of advantages in terms of thinking about uncertainty, making sure all sources of uncertainty, you know, are present in your results. a lot of times that's difficult. If you have a really deep hierarchical model with a lot of different layers in a frequentist model, a lot of times it's hard to really encapsulate all the different sources of uncertainty. So you might be missing some of those sources. A Bayesian model kind of does that naturally. That's one advantage to it. One disadvantage is that Bayesian models are typically very computationally heavy. Modern Bayesian

machineries is a tool called MCMC, Markov chain Monte Carlo. it's a very expensive computational approach, especially as you have a lot of data or very complex models. So that's one disadvantage. And so you definitely have the freedom to pick and choose which approach you want depending on the problem that you're working on.

Question. Yeah, so your first question, the interval, the confidence interval, what it is.

It's because our interpretation of probability, right?

So the frequentest interpretation of probability, if we have a fixed interval, So the parameters fixed, but the fixed interval, if the true parameter is seven, we don't actually know what the parameter is, but it's seven. The probability that it's in this interval that we've constructed this specific interval, five to ten, what's that probability? It's one. It's in the interval. We don't know that it is, but it is. That's the frequentist interpretation of probability.

Whereas a Bayesian says our interpretation of probability is just our current, you know, a distribution that represents our current belief about that. And so it's not this idea of a fixed interval there. For a specific interval, yeah. And so they think about long-run results, like if you got new samples and new samples and new samples, and for each one you made a different interval. The intervals themselves are kind of random. What proportion of those random intervals would cover the true value. Now, that's not a probability in terms of how the frequentists make a probability, but it's sort of similar. They call the idea of confidence. That's this asymptotic view of these intervals covering the truth or not. And then your other question was, why is the Bayesian machinery more computational expensive? Like I said, the modern Bayesian machinery uses a tool called MCMC, Markov chain, Monte Carlo. It's sequential. So at every iteration, you need the value from the previous iteration. It's not parallelizable, right? You have to do things in order to do MCMC. And basically what you're doing is you're sampling from specific distributions iteratively over and over and over and over again. And so that sequential sampling step is what's computationally expensive. It's partially because it's sequential, partially because it's a sampling-based approach. So a frequentist typically won't take sampling-based approaches. They will do more of an optimization-based approach. So we'll talk about maximum likelihood. Maximum likelihood is an optimization problem, right? We have this likelihood function. We're just going to optimize it. That's our best guess. Now, that might not tell us anything about the uncertainty like the sampling approach in the basing case would. but it's a very different procedure computationally. All great questions. Some of this will talk about kind of throughout the quarter like confidence intervals and the Bayesian analog, which is called a credible interval. We won't really get into MCMC. That's more of an advanced topic, but we will talk about and have more simple Bayesian models for the probability distributions that we looked at in our table the other day. So that's the idea of the Bayesian framework, right? We treat these things, treat these population parameters, probability distribution parameters, as random variables, right? We have a distribution that encapsulates our belief about that parameter at any given point in time, typically before we observe the data and after we observe the data. But you could do it sequentially, right? You could collect more data and update again and collect more data and update again. That's another nice feature of the Bayesian framework. Now, there's something called the likelihood. Turns out that both of these frameworks, Frequantists and Bayesian, use this likelihood. They use it in different ways, but they both use the likelihood when doing statistical inference.

All right, so let's go ahead and talk about what this likelihood is. So this is our definition of the likelihood function, L of theta, the likelihood function.

So given a random sample, X1, X2, XN, each of these is a random variable.

Collectively, they make up a random sample, right? Drawn from some population,

right, we know what a population is now, drawn from some population, and that population has some probability density function, or maybe a probability mass function PMF if we're talking about a discrete random variable.

F of X given theta. X is the random variable. Theta is the parameter or maybe parameters, vector of parameters, if we have more than one. Focus on this here.

Random sample. What does that mean? That means these are independent. X1, X2 through XN.

They're independent. And very often you'll see IID. And IID is shorthand for independent and identically distributed.

So in addition to the being independent, right? They're identically distributed, meaning they all have this same probability density function.

We don't have x1 is normal and x2 is binomial and all of that, right? They all come from the same distribution. They all have the same parameters, right? And that helps us kind of identify something about these parameters, right? We have some variation, multiple data points informing a parameter.

Okay, so we have a random sample, x1 through xN, drawn from some population, with PDF or PMF, F of X given theta. Or at least we model the population using that. Then the likelihood function, L of theta, notice it's a function of theta, but when we talked about probability density functions or probability mass functions, we view them as a function of X given theta, right? But this likelihood function is going to be a function of theta. And it's defined in this way. It's a product over all n of these observations from I equals 1 to n of their probability density function. And they all have the same F, same theta. It's just the value of the random variable that's different across it. Now, in this definition, this is only for a random sample, independent. You can construct likelihoods for non-independent data. It's more complicated, not something we're going to cover, but basically you lose this product feature. This product is coming from this independence assumption. Remember from probability, if you have two independent random variables, their joint probability distribution is the product of the two. It's that idea, right, being fed into our likelihood function. And as I said, F of X given theta, when you work with this as a probability density function or probability mass function, right, you're conditioning on theta, and X is kind of the thing, the input to this function, but in statistics, we observe the data, we don't know the value of the parameters, right? And so even though we write it in this way, the input to the function is theta. and we actually know X, so we're sort of conditioning on X when we view this function. We've collected some data, or maybe we're making a general rule after we collect data, right? And we view this as a function of theta. And so it kind of tells us, right, if L of theta is 5, it's equal to 5 for the value of theta at 1, and it's equal to 500 for the value of theta at 2, right? It tells us that in some sense, two is a much more likely value than one, right? And I say that very loosely without defining what we mean by more likely, because that's going to depend on the framework that we're using, so we'll get into that in more detail. But it's a function of theta that tells us which values are kind of more likely than the other in some sense. Yeah, question. If we had data that was, let's say, like, a burden of redistribution, but then let's say like the parameter was like a beta distribution, for the likelihood we would only multiply the probability, is the parameterly distributed? it was only concerned with the data based on the triangle? Yes. And also notice that you said the distribution for the parameter is beta. Only the Bayesian would have a distribution for the parameter. The frequency wouldn't at all. The frequency would only have that binomial piece. But both frameworks would use the likelihood of it. Yeah, there's nothing about this that treats theta as random. Right. It's just some function of theta, but it's not necessarily saying that theta is random. It's just some function of theta that we're going to use to, you know, estimate theta.

And the way they do that is a little bit different. Here I'm just kind of saying

that same thing, even though the mathematical expression is the same as what you talk about in probability theory, right? We're viewing this as a function of theta, the unknown parameter, right?

The data X is what we're going to observe. Now we look back at our table from our review last time, and I brought up this idea of parameters having constraints. The parameter space has a constraint, we can use what's called an indicator function to signify this in the likelihood. Okay, so let me kind of talk about what an indicator function is real quick.

So an indicator function we label with this I, and then in parentheses will have some expression, maybe something like zero less than or equal to p, less than equal to one.

This is an indicator function, right?

An indicator function only returns the values one or zero. It returns the value one if that expression is true, and it returns the value zero if that expression is false. So this indicator function here that we've written would return one if zero is less than equal to p, less than equal to one, and it would return zero otherwise.

I'll go over an example of how this might look in a likelihood in a moment, but just be aware that that's what these indicator functions are, right?

We use them to identify constraints for our parameter as we're working with different likelihoods if there is such a constraint.

Okay, so let's say you have 10 observed independent flips of a coin, and eight of those flips are heads. 10 independent flips, observed eight heads. And we'd like to infer something about the true probability of heads, which is P. We don't know what that is. We'd like to learn something about it. We've collected data, but we don't know the true probability of heads. So let's write a likelihood function for this problem. So the likelihood function is going to be a function of our parameter, our unknown parameter, P in this case. and it just comes from this equation in our definition. We know each of these coin flips is a Bernoulli trial, a Bernoulli distribution. And there's 10 of them. So we have a product over all of our observations from I equals 1 to 10, 10 coin flips. And each of these is Bernoulli. And the Bernoulli probability mass function, or PMS, PMF, since this is a discrete random variable, it is going to be p to the xLI, our random variable, the Ith observation, times one, minus p to the one minus x i so if x is one we've observed heads right that probability mass function evaluates to p if we've observed tails x is zero that probability mass function evaluates to one minus p and then we're going to do this indicator function so the indicator that p is between zero and one so why do we do this well now notice that if we input any value of p between zero and one we're going to return this piece right it's going to tell us how likely p is for for that specific value of p. But if we get, if we input some value of p that's not allowed, like two, then rather than returning to the product of two to the x times one minus two to the one minus x, it's just going to return zero because this indicator function evaluates to zero. So any value that's not allowed, the likelihood is zero, specifically because this indicator function evaluates to zero. So this indicator function is making sure that we're respecting the parameter constraints for this specific model, this Bernoulli model. It's a function of all of the data points, right? In this specific case, we can see it is going to simplify based on the number of heads. So for all of the ones that are heads, like I said, if X is one, X is heads, this returns P to the 1 times 1 minus p to the 0, which is just p, right? There's 8 with heads, so this is going to simplify to p to the 8th power, 1 minus p. So this is like a more general rule for this process of flipping 10 coins. For the process of flipping 10 coins, this is the likelihood. And then for the specific case where we observed eight heads in this indicator function, for the specific case where we've observed eight heads, this is the likelihood function. For this specific case, we can tell which values are more likely than others in some sense of likely.

Here we treated these as 10 independent renewly flips. Now if you remember from probability theory, If you have repeated independent Bernoulli draws from the same distribution, that's what distribution? What was the second one? Binomial. Yeah, if we repeat independent Bernoulli trials, we get a binomial distribution. So another way we could have looked at this problem is instead of 10 Bernoulli random variables, one binomial random variable, parameter N is equal to 10. Right. So maybe let's say now YI sum of XI. YI is now a binomial. random variable.

We could write the likelihood function as a function of p product from i equals 1 to 1. There's only one term here because we have one binomial random variable.

And the binomial distribution has probability mass function and choose x or y i in this case.

It's 1 minus p and n minus binomial probability mass function. And then in this specific case, oops, make sure to write my indicator function here.

Indicator means 0 and 1. And in this specific case, this evaluates to 10, choose 8, p the 8, times 1 minus p squared, times our indicator function.

Okay, looks very similar what we had here.

The only difference is this term 10 choose 8. But notice 10 choose 8 is a constant. It's not a function of p.

So it's not really changing the shape of this likelihood function. It's just shifting things up or down.

It's a constant. One value, p is more likely than that. and another under the first. It will also be more likely than the other under the second. Yeah, question. Compare this likelihood. So this is not a distribution. Yeah, that's something I want to point out. A likelihood function is not a distribution, right? It's not a distribution over P. That's why frequentists are willing to use it. They don't want to assume a probability distribution for the parameter. This is not one. Why isn't it one? Well, it doesn't integrate to one. We know a probability distribution has to integrate to one. If you integrate this function from P, 0 to 1, you will not get 1. It's not a probability distribution. It doesn't meet that criteria. It just tells us in some sense which values more likely than the other than another.

It's outputting some notion of how likely a parameter theta is given the data we observed.

So if we look at this plot here, this is that likelihood function for those 10 coin flips with eight heads.

I'm plotting the second version here, but either one.

So I'm plotting this function here from 0 to 1. This is the likelihood function. It tells us which values of p along the x-axis here are more or less likely, given the data we observe. We observe eight heads out of 10, right? So the probability of heads being called 0.1 seems very, very unlikely, and that's what we observe here. It's a very, very low value, very close to 0 for p equals 0.1. Now p equals 0.8, very, very likely. Intuitively, that makes sense. If we got 8 out of 10 heads, probability of 0.8 seems like a pretty low. likely value for heads.

Does that make sense? The binomial variable you have like the two times of figure, but that isn't matter because it's just kind of like a, it's just sort of stretching the function that is like still the same. Or still the same kind of like you would at the base is the same? Yeah, whether or not you, you know, divide by this 10 choose 8 or include it here, right? This is still going to be the highest value. These values are just going to be low and so on. Okay, so a

coin flip is modeled as Bernoulli trial with probability of heads equal to p. If P is a probability, what restrictions do we have on P? The coin can be head or tails, but what's the restrictions on a probability? What values can a probability take? Yeah, it can only be between 0 and 1. So that's our parameter space. That's the restrictions on our parameter. And so here, this is saying for any value of P that doesn't meet our restrictions, this likelihood function is going to return 0. There's zero likelihood that P is negative 3. There's zero likelihood that P is 5. But there's some positive likelihood that p is a value between 0 and 1. In the first line or the second one. So in the first line, notice that we have eight heads. For all of those eight heads, right, we have p to the 1 times 1 minus p to the 0, which is just p. So for the eight heads, the product of those is p to the 8. And for the two tails, the product of these is 1 minus p squared. In the second case, we said it was a binomial random variable. there's only one of those binomial random variables, right? And so the product is just over one term. So this is that one term. Is there one more question? Yeah, back here. Yeah, you can update the likelihood function. What you do with the likelihood function after that, whether or not it's valid to update that process as well as going to depend on what that process is. But you can certainly, you know, if you flip the next coin and you get heads, you could update this to 11, choose 9, p to the 9, 1 minus p squared. There's a question back here. When we extend Bayes theorem and talk about for probability distributions rather than just a probability, yeah, it will be the probability of the data given the parameter. Yeah, but we haven't talked about Bayes theorem for this yet. Right now, we're just defining what the likelihood function is. So we haven't committed to the Bayesian or frequentist case yet. We're just saying this is a likelihood function. We'll talk about what to do with it in Bay's rule, mostly Thursday, but for now it's just a function of p, of theta. Yeah, because our probability, the probability distribution we're using to model the data is Bernoulli, and the Bernoulli parameter, which we call P, has the restriction of being between zero and one. Our indicator has to reflect that, that specific parameter restriction.

If we were modeling something as poisson, right, the poisson parameter lambda has to be positive, greater than zero. So there we would have an indicator that lambda is greater than zero, right? So it's just reflecting whatever parameter restrictions we have for a given probability distribution.

And again, the likelihood function is not a probability distribution. It does not integrate to one. In fact, you should go ahead and try this. Take this function, integrate out P from zero to one. You won't get one. So it's not a probability distribution. But it does give us some sense of which parameter values might be more likely to generate the data that we happen to see. If we happen to see eight out of ten coins return heads, right? These values are much more likely than these values over here. That's kind of the idea between the likelihood function. We're not really doing anything with it yet. We're just saying general notion of something is more likely than the other.

A couple more examples. So let's say that scores from some specific personality test are normally distributed, and that normal distribution has a mean mu. We don't know mu, and it has a standard deviation. 10, let's say for now we do know the standard deviation. And so there's really only one unknown parameter here, even though the normal distribution has two parameters. So mu is unknown. We'd like to learn something about it. We randomly select three individuals.

We have a sample size of three, and we measure their scores. Let's say we get 45, 55, and 50.

That tells us something about this parameter mu. Let's go ahead and write the likelihood function for this problem.

We go back to our definition of the likelihood function. We know we're going to have a product because it's a random sample. We're going to have a product, and it's going to be three because we have a sample size of three. And we need to know what this function F is, right? F is our probability mass function, our

probability density function. Well, we were told that these are normally distributed. So F is going to be the normal PDF, normal probability density function. So you could go ahead and look that up in that table. In this case, XI, right? I could be one, two, or three. For any of these, it doesn't matter. given our parameter, I'm going to write mu here instead of theta because we're calling our parameter mu.

This probability density function is the normal probability density function, which is one over square root two times pi times sigma squared.

Sigma squared is the variance of the normal distribution. We weren't given the variance, right? We were given the standard deviation, but we know that the variance is the square of the standard deviation or 10 squared 100 times 100.

Okay, so we have that. And then, times e to the negative x minus the mean, mu squared, over two times the variance. Is everyone familiar with this notation, XG? So this is the normal probability density function when the variance is 100 or the standard deviation is 10. So this is the PDF of one single observation. So now when we go to write our likelihood function for mu, L, mu, it's going to be a product over For our three observations, product from i equals 1, 3, f, xI, given mu, which we just defined. The product, I equals 1, 3, 1 over the square root. It's pi, it's 100, it's e to the negative xI minus mu squared, 100. Now, since we have a product of these three exponential terms, right, we know that they collect addatively inside the exponent, so we can simplify this a bit. three of these constant terms multiplied together, or 1 over square root of 2 times pi times 100, the third power. Then we have e to the negative sum now, equals 1 to 3 xi minus mu squared over 200. Yeah, question. Yeah, good question. In this case, mu doesn't have any parameter restrictions, right? It can be any value between negative infinity and infinity. You could write an indicator function, but mu is between negative infinity and infinity, but it will always return one. So in this case, we don't actually need to write it. If it helps you to write it in that case to remember to write them, go ahead. But if there's no restriction, it's not strictly necessary. It's strictly necessary when there is a restriction on the parameter. Yep, we haven't talked about maximum likelihood yet. We will in a couple weeks, but whether or not we multiply by this 10 to the 10 choose 8 term here, the maximum is still going to be here. So you can ignore constants. And we'll talk about that when we talk about maximum likelihood. And then the last thing here is we have actual values for x, right? We were told that x1 was 45, x2 was 55, x3 was 50, so we can go ahead and plug those in as well. And so for the specific data we observed, this likelihood function is going to be one of the square root of 2 times pi times 100 to the third power. E, the negative 45 minus mu squared, it's 55 minus mu squared, it's 50 minus mu squared, it's 50 minus you squared. squared all over 200.

This is the likelihood function for me, given the data, the three data points that we observed. You could plot this if you wanted to. For right now, we're just trying to make sure we know how to write down a likelihood function for a given problem. We're not really doing anything else with it yet.

That's the likelihood function. Again, both frequentists and Bayesian's use the likelihood function. But now let's talk about Bayesian inference a little bit. We're just going to touch on it today. We'll really get into it more on Thursday. So let's suppose we have some statistical model, a probability distribution for our data, with parameter theta, and again, maybe theta is a vector, maybe we have more than one parameter. If we're willing to be basian, we're willing to treat theta as random, then the distribution that we assign to it, assigned to theta before observing any of the data, is what we call prior distribution.

And this notation is what we'll use for the prior distribution, pi of theta. Pi of theta is just symbolic for the prior distribution for our parameter Theta. It could be either a PDF or a PMF depending on whether this parameter is continuous or discrete. Most cases, parameters are continuous, but they don't necessarily have to. Look at an example. Let Theta denote the probability of obtaining a

head when some coin is tossed. And the only thing we know about this coin is that it's either fair or it has a head on each side. That's all we know about the coin. So what's an appropriate prior distribution for this case? 0.75 what? So our parameter is the probability of obtaining heads. So we'd like to put prior on that parameter, on the probability of obtaining heads. B Bernoulli? Yeah. The right? It's discrete, right? We only have two choices. There's only two possible values for theta. Only possible values for theta, or our parameter space. The only two possible values are 0.5, if it's a fair coin, or one, if it has heads in both sides. We can't have a probability of 0.75, because it's either a fair coin or it has heads on both sides. These are the only two possible values. So our prior should reflect that. Our prior should have the form like this, Pi Theta, prior probability mass function in this case, PMF because we're talking about something that's discrete. It should look like something A, some value A for Theta equals 1, some other value B for Theta equals 0.5, and it should return zero otherwise. And I will often just put OW for otherwise, by the way. Okay, so it should look like this where it's a valid probability distribution, right, PMF. So A plus B should sum to one, and A and B they both be greater than equal to zero. So something that looks like this is an appropriate prior distribution. So one example might be this. Maybe before collecting any data, we think, oh, there's probably a fairly small probability, right, that it has heads on both sides. Coins aren't usually like that. So maybe only 0.2 probability that theta's one. And we probably think there's a larger probability that it's a fair coin because most coins that we come across are fair. So 0.8 and 0 otherwise. This is a reasonable prior for this case. It reflects the fact that our parameter is discrete in this case. It only has two possible values. So it reflects our parameter restrictions. And maybe it reflects our prior knowledge that fair coins are more common than fair coin. It doesn't have to be 0.2 and point A. You could pick your own values, but something along these lines is a reasonable prior distribution. Similar example now, let's say that theta represents the proportion of defective iPhones in an Apple warehouse. So now our parameter space is the interval, zero to one. Proportions have to be between zero and one, but it's now continuous instead of discrete. And let's say we don't really have any prior knowledge. We have absolutely no idea what a reasonable value for theta is other than that it has to be in this interval. It has to meet our restriction. In that case, we might choose a uniform prior on this interval. Pi theta, prior distribution. In this case, we're talking about a continuous random variable. It's a proportion. So this probability density function now is uniform. And the uniform distribution is 1 for any value between 0 and 1 and 0 otherwise. So this is the probability density function. You could even plot this. Let's say, let's say our x-axis represents theta, or y-axis represents the PDF. pi of theta, right?

What would this look like? Well, we'd have a value of one for anything between zero and one, and we'd have a value of zero for anything outside of it.

This is our probability density function for this example, where we put a uniform prior on this proportion.

So a uniform prior can be a great type of friar when you have a bounded parameter like this, and you don't really have any prior information to go off of. In other cases, you might have prior information. Maybe you're working with a scientist, right, and they know something about a parameter, or maybe you have prior experience to build off of. In this case, you might want to construct a prior with a specific standard deviation, low uncertainty, reflect your prior knowledge.

So let's look at an example of that.

Suppose that the lifetime and hours of some fluorescent lamps, there to be observed, and the lifetime of any particular lamp has an exponential distribution with parameter theta. So we have a probability distribution for the data, these lifetimes, that probability distribution has a parameter theta.

We don't know what theta is. is it's unknown, right?

But maybe we do have some prior knowledge. And that prior knowledge tells us we're going to use a gamma prior distribution with a specific mean. We want the mean to be 0001.

0.0002 and we want the standard deviation to be 0. Whatever our prior knowledge is, it tells us 002 is a reasonable value.

we think 0. We'll center our prior around that value. And we'll have relatively little uncertainty, a pretty small prior standard deviation. So we want to write pi of theta, our probability density function now for this problem. We know it's going to be a gamma distribution, right? But what are the specific parameters of that gamma distribution? This is the probability density function for the gamma, cause these parameters beta and alpha, but beta and alpha are not the mean or the standard deviation. We're given the mean and the standard deviation, right? How do we find the appropriate beta and alpha to reflect our prior and alpha? We're going to have some gamma prior distribution. It's going to have parameters. I'm going to call them alpha 0 and beta 0, right, to reflect the fact that these are our prior. If it was our posterior later after we collected our data, maybe I'd label them alpha 1, beta 1. You can label them whatever you want. But we want to find out what these parameters are. What do we know about the gamma distribution? Well, we know the mean. If we didn't know the mean, we could look it up from our table. The mean is alpha over beta, and we know the variance, or we could look it up from our table. look up the variance, and it's alpha over beta squared. So now we have something that translates from what we are given, the mean and the variance, into what we'd like, which is the parameters of this prior distribution. So in our case, the mean, alpha 0 over beta 0, we want that to be 0.002. We're talking about the standard deviation, not the variance. The standard deviation is the square root of the variance. So the standard deviation, or square root of alpha 0 over, 001.

beta 0, that's 0. The problem statement. Now we've got two equations, two unknowns, system of two equations, we can solve for alpha-0 and beta-0, and get the appropriate prior distribution for this problem.

Okay, so we'll go ahead and do that. Solving, okay, take this equation, multiply both sides by beta-0, square root of alpha-0 equal to 0.000-0-0-0-0-0-0 equal to 0.001 squared, beta 0 squared. Now let's go ahead and take this and plug it into our first equation. So then we have 0.001 squared times beta squared over beta equal to 0.002, or solving for beta 001 squared, which is 20,000.

0 to 0.0 0.02 over 0. Okay, so now we know the value of beta, or beta 0.0, it's 20,000.

Take that now, plug it back into equation one and solve for alpha 0. Alpha 0 over 20,000 is equal 0.

to And that gives us alpha We know in this case our prior distribution for this parameter is going to be a gamma distribution parameters for 20,000.

And we know what the probability density function for a gamma distribution is so we could actually write out the PDF for this prior. It's just this gamma distribution as a function of theta, plugging in the alpha and beta that we solved. Theta, prior distribution, probability density function for theta, beta to the alpha, 20,000, the fourth power, over gamma of alpha. Remember, gamma of an integer is that integer minus one factorial, or three factorial in this case. And then we have X to the alpha minus one. In this case, we're talking about theta, right? It's a distribution of theta, not x. So we have theta to the alpha minus one, or theta to the third power. And then we have E to the negative x beta, or in our case, E to the negative theta times beta. Theta is our random variable. E is a negative 20,000 times theta. So just to kind of recap here on prior distributions, we learned some important rules here, one. Pay attention to whether your parameter is discrete or continuous. Your prior should reflect that. We saw in the case where you don't really have any prior information to go

off of, right, in cases where a uniform distribution makes sense, right? That can be a nice prior that doesn't reflect any of your own beliefs, right? It's just uniform, it's flat. In other cases, you might have prior information to go off of. Typically, we think about information in terms of like center and spread or mean and variance, right? And so we saw for a distribution like the gamma that's not directly parameterized by the mean and the variance or the mean and the standard. deviation, how we can use our own prior knowledge of the mean and the standard deviation to find the appropriate values for such a distribution. Be questions. Last, it mentioned like an exponential distribution, I think, was it for the data, but did that have any role in influencing how we came up with our effect? Yeah, so we said the exponential distribution for the data is the model we assumed for the data, but our prior doesn't use the data, so we didn't need that information there. Later, when we talk about the posterior distribution, when we update our business, we update our beliefs after observing the data, we would use that, right? We'd use both the prior and the likelihood or the distribution of the data to get some type of posterior distribution, which is what we'll talk about Thursday. So as I just said, we'll talk about the posterior distribution Thursday, which actually gives us a way to do Bayesian inference now. And again, I'll stick around for a bit if there's any questions.